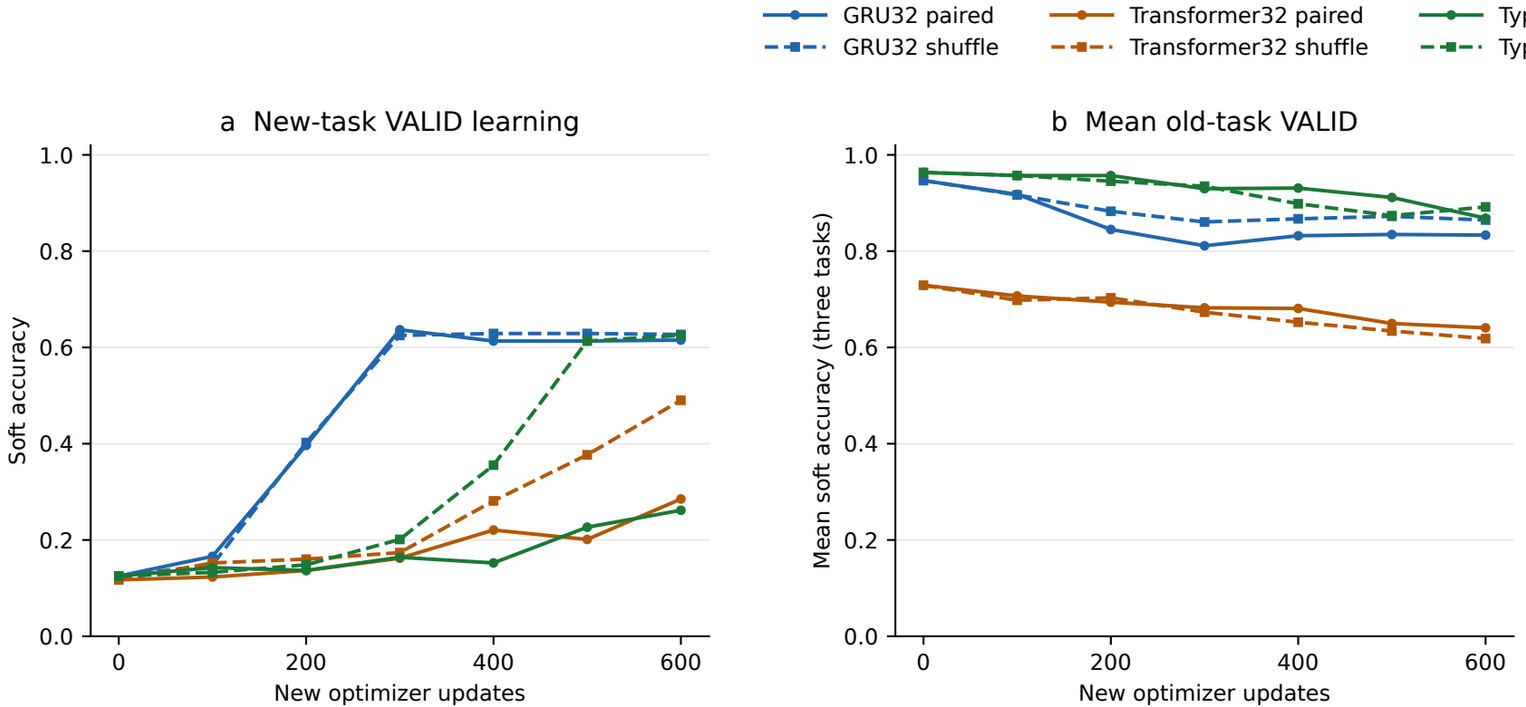
Paired minibatches versus the same globally shuffled TRAIN pool



c Parameters and measured cost scopes

Model / batch	P (k)	Train (s)	Science (s)	Cold (s)	Eager (ms)
GRU32/pair	14.024	4.84	10.32	16.26	1.65
GRU32/shuffle	14.024	5.18	10.39	16.17	1.64
TF32/pair	26.760	6.01	11.05	16.89	1.08
TF32/shuffle	26.760	6.95	11.92	18.04	1.06
Typed/pair	10.991	11.74	31.71	37.91	42.04
Typed/shuffle	10.991	11.71	32.33	39.77	42.57

Eager: B64, max L39; soft wall p50, 20 timed + 3 warmups.
Train / science / cold are nested scopes: do not sum.
NN / Adam: typed 44.0 / 88.0 kB; GRU 56.1 / 112.2 kB;
TF 107.0 / 214.3 kB. Replay 55.3 kB per tail.
Typed training uses one Graph; eager inference does not.

Development seed0 only; no Primary or architecture-advantage claim. Each tail restores its own model's same stage2 NN / Adam / RNG / Replay.
Current-row multiplicities and replay draws match; grouping and temporal order both change. Initial old-task competence differs across model families.
Six new tails: 3,600 updates, 46.43s training / 107.72s science / 145.04s cold. One-time formation + CPU admissions bring cold total to 168.15s.
Earlier training, configurations and failed attempts remain charged separately. Full training FLOPs, GPU busy time, energy and provider-effective compute are unknown.